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Process simulation of methanol production via carbon dioxide hydrogenation

Jinfeng Fu [a,*], Mohammed Sh Majid [b], Farag M.A. Altalbawy [c,d], Radhwan M. Hussein [e], Ibrahem Waleed [f], Ibrahim Mourad Mohammed [g], Rahman S. Zabibah [h], Kadhum Al-Majdi [i], Abdul Malik [j]

[a] School of Chemical and Environmental Engineering, Jiaozuo University, Jiaozuo, Henan, 454000, China
[b] Computer Techniques Engineering Department, Al-Mustaqbal University College, Babylon, 51001, Iraq
[c] National Institute of Laser Enhanced Sciences (NILES), University of Cairo, Giza, 12613, Egypt
[d] Department of Chemistry, University College of Duba, University of Tabuk, Tabuk, Saudi Arabia
[e] College of Pharmacy, Ahl Al Bayt University, Kerbala, Iraq
[f] Medical Technical College, Al-Farahidi University, Iraq
[g] Al-Nisour University College, Baghdad, Iraq
[h] Medical Laboratory Technology Department, College of Medical Technology, The Islamic University, Najaf, Iraq
[i] Department of Biomedical Engineering, Ashur University College, Baghdad, Iraq
[j] Department of Pharmaceutics, College of Pharmacy, King Saud University, Saudi Arabia

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ABSTRACT

The performance of a packed bed reactor for CO_2 conversion to methanol was investigated. Because of the exothermic reaction, temperature and concentration gradients happen along the reactor. A 2D packed-bed reactor model was applied for numerical investigations, coupled with detailed reaction kinetics and transport phenomena. The effect of feed inlet pressure and temperature on CO_2 conversion and temperature profiles along the reactor was evaluated. The results showed that firstly increase in the temperature in the reactor because of the presence of exothermic reactions and then slight decrease in the temperature at top section of the reactor. Pressure was decrease from 55 bar to around 54 bar along the reactor. It was found that the CO_2 conversion at outlet of reactor was 18.18 % at T = 475 K and it was increased to 23 % at temperature of 498 K and 555 K. The CO_2 conversion was reached its maximum at short distance from the reactor entrance at T = 555 K. Also, the outlet temperature of gas mixture was increased from 522.53 K to 534.56 K with increasing feed inlet pressure from 35 bar to 55 bar. The CO_2 conversion was 18.12 %, 20.77 %, and 22.96 % at pressures equal to 35, 45, and 55 bar respectively.

1. Introduction

Emissions of carbon dioxide, which are mostly produced by heavy industries and traditional power plants, constitute a danger to the environment and significantly contribute to global warming. Effective actions to reduce climate change and manage air pollution are urgently needed as a result of rising CO_2 levels in the atmosphere [1]. About 81 % of the total amount of greenhouse gases comes from CO_2 [2]. One novel strategy uses CO_2 as a chemical feedstock to produce methanol [3]. Methanol is an important C_1 chemical that is used to produce a variety of other compounds, such as formaldehyde, methyl *tert*-butyl ether, olefin, and others [4]. Methanol

* Corresponding author.
E-mail address: 15239126088@139.com (J. Fu).

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has potential as a car fuel and is less harmful to the environment than conventional fossil fuels [5]. Methanol production capacity is now estimated to be over 100 million tonnes per year, and it is predicted to rise by 10 % per year [6].

There are several methods for methanol production on commercial scale. The common method for methanol production is performed through multiple main steps: generation of syngas from coal gasification or natural gas reforming; conditioning of syngas; and methanol synthesis and purification. Around 80 % of methanol is made using steam reforming technology [7]. In recent years, methanol generation in Kraft pulp mills has drawn a lot of attention. It provides some advantages, including sustainable development, less waste, and revenue stream diversification [8]. Another technique to produce methanol is CO_2 hydrogenation. Green hydrogen combined with CO_2 hydrogenation to methanol offers a workable process approach that lowers CO_2 emissions while increasing the ability to use renewable energy [9]. CO_2 Hydrogenation by ZrO_2/Cu(111) was investigated and it was found that ZrO_2/Cu(111) is better than ZnO/Cu(111) due to the different sizes of the oxide islands as well as in the strength of oxide-metal bonds [10].

Due to the chemical stability and inertness of carbon dioxide, an effective catalyst is one of the essential components in the CO_2 hydrogenation to methanol process. The selectivity of methanol and the rate of carbon dioxide conversion are significantly influenced by the catalysts. In general, three categories may be made out of the catalysts depending on the different active ingredients. Cu-base catalysts including $Cu/ZnO/Al_2O_3$ and $Cu/ZnO/ZrO_2$ are the most common catalytic systems used for carbon dioxide hydrogenation [9,11]. Conventional coprecipitation, reduction-deposition, lame-combustion, or parallel-slurry-mixing methods are used for the preparation of the Cu-base catalysts [12]. Fang et al. [13] used $Cu/ZnO/Al_2O_3$ catalyst for dehydrogenation of 2-butanol. Pd-modified (Pd-modified) $CuO–ZnO–Al_2O_3$ catalysts were used for methanol synthesis from CO_2 hydrogenation and it was found that Pd enhanced the CZA catalyst's performance at low temperatures [14]. Plasma-catalytic CO_2 hydrogenation to methanol was performed using CuO–MgO/Beta catalyst. The conversion and selectivity of CO_2 hydrogenation to methanol was obtained 8.5 % and 72 % [15]. Different aspects of CO_2 hydrogenation to methanol over various catalysts have been investigated recently [16–20]. However, it should be mentioned that the carbon dioxide conversion ranges from 1.5 % to 25.9 % because of the thermodynamic equilibrium limit [21] and it is a high-pressure system. Also, the production of water during the reaction can cause the inactivation of the catalyst [22]. Heracleous et al. [23] investigated the use of zeolite 13X as a water adsorbent in CO_2 hydrogenation reactions to increase the performance of processes. It led to a significant enhancement in the carbon dioxide conversion and methanol yield (115 %).

Due to the importance of the carbon dioxide hydrogenation process, the theoretical approach could provide valuable information to understand and optimise the process [24]. In recent years, mostly methanol steam reforming was modelled and simulated. Modelling and simulation of the H_2 production and CH_3OH steam reforming in a double-jacketed membrane reactor were performed. H_2 recovery, methanol conversion, and production rate were better in a double-jacketed membrane reactor than in an auto-thermal reactor [25,26]. CH_3OH production using carbon dioxide hydrogenation was theoretically investigated with zeolite's catalyst in a pressure swing adsorption reactor. It was proposed that the highest conversion can be obtained at the temperature and pressure of 250 °C and 50 bar in a step. The selectivity and productivity were 83–96 % and 0.023–0.13 mol m^3 s^{-1} [27]. There are few research studies to investigate methanol production by carbon dioxide hydrogenation in more detail. Three methanol reactor configurations were simulated to predict the process performance. It was observed that axial dispersion does not play a significant role and it was ignored and the reactor with recycled showed the best performance [28]. Leonzio and Foscolo [29] studied methanol production the structured as well as non-structured catalyst packing and it was found the carbon dioxide selectivity to CH_3OH and methanol yield were 28 % and 4 % higher in structured packing. A 1D model was used to study CO_2 hydrogenation in a fixed-bed reactor. Maximum methanol yield is observed to be limited by 1D solution [30]. As can be seen, most of the developed models are 1D, therefore, it will be valuable to investigate the system in 2D as it is possible to get more accurate results. The important point of this study is focusing on the analysis of the system performance for a wide range of operating conditions. Also, previously developed model, the hydrogenation of carbon monoxide was not considered in the kinetic of the developed model [29], however, the model in the current study considers this reaction which can improve the accuracy of the model.

In the current study, the emphasis is on the development of a 2D comprehensive mathematical model to predict the synthesis of methanol by catalyst ($CuO/ZnO/Al_2O_3$) in the multi-tubular reactor. The effect of different operating parameters on methanol production in the reactor was investigated.

2. Model development

Fig. 1 represents the domains of Lurgi type multi-tubular reactor. The gas mixture enters from the bottom of the reactor into the tube side while the boiling water passes through the shell side of the reactor counter-currently. The tube side of the reactor is packed with $CuO/ZnO/Al_2O_3$ catalyst [29].

The kinetic mechanism considered in the current work was suggested by some researchers previously. It was suggested the hydrogenation of CO_2 and CO disregarding the water-gas shift [31,32]. In the current study, 3 reactions including hydrogenation of CO_2 as well as CO, and reverse water-gas shift were considered as follows [33,34]:

$$CO + 2H_2 \leftrightarrow CH_3OH \quad \Delta H^0_{298} = -90.55\frac{kJ}{mol} \tag{1}$$

$$CO_2 + H_2 \leftrightarrow CO + H_2O \quad \Delta H^0_{298} = 41.12\frac{kJ}{mol} \tag{2}$$

$$CO_2 + 3H_2 \leftrightarrow CH_3OH + H_2O \quad \Delta H^0_{298} = -49.43\frac{kJ}{mol} \tag{3}$$

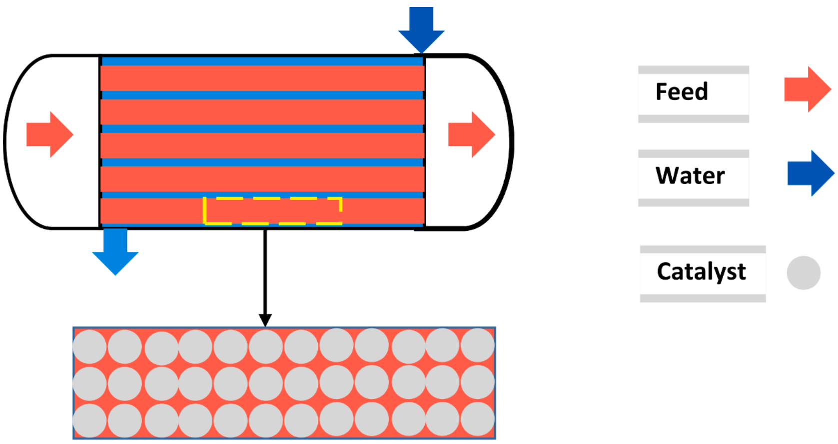

Fig. 1. Lurgi type multi-tubular reactor for CO_2 hydrogenation.

It should be noted that side reactions and by-products were ignored as they are quantitatively negligible [35]. The reaction rate for the reactions of (1), (2), and (3) can be written as follows [34]:

$$r_1 = \frac{k_1 K_{CO}\left(p_{CO}p_{H2}^{1.5} - p_{CH3OH}/\left(p_{H2}^{0.5}K_{p1}^{0}\right)\right)}{\left(1 + K_{CO}p_{CO} + K_{CO2}p_{CO2}\right)\left(p_{H2}^{0.5} + \left(K_{H2O}/K_{H2}\right)^{0.5}p_{H_2O}\right)}\rho \tag{4}$$

$$r_2 = \frac{k_2 K_{CO2}\left(p_{CO2}p_{H_2} - p_{H2O}p_{CO}/\left(K_{p2}^{0}\right)\right)}{\left(1 + K_{CO}p_{CO} + K_{CO2}p_{CO2}\right)\left(p_{H2}^{0.5} + \left(K_{H2O}/K_{H2}\right)^{0.5}p_{H_2O}\right)}\rho \tag{5}$$

$$r_3 = \frac{k_3 K_{CO2}\left(p_{CO2}p_{H2}^{1.5} - p_{CH3OH}p_{H2O}/\left(p_{H2}^{1.5}K_{p3}^{0}\right)\right)}{\left(1 + K_{CO}p_{CO} + K_{CO2}p_{CO2}\right)\left(p_{H2}^{0.5} + \left(K_{H2O}/K_{H2}\right)^{0.5}p_{H_2O}\right)}\rho \tag{6}$$

Kinetic constants in the reactions are provided in Table 1.

The proposed model was simplified according to the assumptions that follow [28,29].

- Axial dissuion and catalyst deactivation is negligible.
- The components were assumed as ideal gas
- A steady temperature and pressure distribution inside the catalyst particles

Table 1
Kinetic constants of methanol production by CO_2 hydrogenation [34,36].

Rate constants		
k = Aexp(B/RT)	A [mol/s.kg.bar]	B [J/mol]
k1	4.89×10^{7}	−113000
k2	9.64×10^{11}	−152900
k3	1.09×10^{5}	−87500
Equilibrium constants		
$Log_{10}K$ = A/T-B	A [K]	B [−]
$K_{eq,1}$	5139	12.61
$K_{eq,2}$	−2073	−2.029
$K_{eq,3}$	3066	10.5
Adsorption equilibrium constants		
k = Aexp(B/RT)	A (1/bar)	B [J/mol]
K_{CO}	2.16×10^{-5}	46800
K_{CO2}	7.05×10^{-7}	61700
$(K_{H2O}/K_{H2})^{0.5}$	6.37×10^{-9}	84000

The mass transfer of components in the packed reactor is determined using the following equation [28]:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C_i}{\partial r}\right)-\frac{U_z r_{tot}^2}{L_{tot}De}\frac{\partial C_i}{\partial z}+\frac{r_i}{r_{i,o}}\frac{r_{i,o}r_{tot}^2}{DrC_{in,CO2}}=0 \tag{7}$$

where r and z denote radial and axial coordinates, Ci refers to the molar concentration of component i, and Uz and r_{tot} are the superficial gas velocity and the radius of the tube in the reactor. The symbol Dr means the dispersion coefficient in r-direction. In addition, r_i is the reaction rate of i component in mol/m^3s. The superficial gas velocity was calculated based on the following equation [28]:

$$U_z=U_{in}\frac{P_{in}}{P}\frac{T}{T_{in}}\frac{\sum_i F_i}{F_{tot,in}} \tag{8}$$

The heat transfer equation in the reactor is presented as follows [29]:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)-\sum_{j=1}^{2}\frac{\Delta H_j Dr_z C_{in,CO2}}{KeT_{in}}\frac{r_{tot}^2 R_{i,o}}{DeC_{in,CO2}}\frac{R_{i,o}}{R_i}-\frac{U_z\sum C_{p,i}C_i r_{tot}^2}{L_{tot}Ke}\frac{\partial T}{\partial z}=0 \tag{9}$$

The T is the temperature, ΔHj denotes the heat of the reaction. Dr and Ke are calculated using the following equations [29]:

$$K_e=K_g\left(\frac{\varepsilon}{1.5}+\frac{1-\varepsilon}{0.312\varepsilon^{2.32}+\frac{K_g}{K_{cat}}\frac{2}{3}}\right) \tag{10}$$

$$K_g=\left(0.01234\times10^{-3}\right)+1.84375\times10^{-7}T \tag{11}$$

$$D_r=\frac{\frac{d_p\mu}{8}}{\left(1+19.4\left(\frac{d_p}{d_t}\right)^2\right)} \tag{12}$$

$$\frac{P_{in}}{L_{tot}}\frac{\partial P}{\partial z}=-\frac{G}{\rho g_c d_p}\left(\frac{1-\varepsilon}{\varepsilon^3}\right)\left[\frac{150\,(1-\varepsilon)\,\mu}{d_p}+1.75G\right] \tag{13}$$

Pad is the dimensionless pressure, Pad ¼ P/Pin, Pin is the feed pressure at the entrance of the reactor in Pa, G refers velocity symbol in $kgs^{-1}m^{-2}$, ρ defines the density of gas in kg/m^3. Boundary conditions for the developed model are provided in Table 2. Furthermore, a list of the values of parameters used in the developed model is given in Table 3.

3. Results and discussion

The molar flow of components as a function of the length of the reactor is shown in Fig. 2. The results are quite similar to other previous studies [28,29]. The carbon monoxide content in the feed was 0.07 mol/s. According to Fig. 2, as expected, the molar flow rate of hydrogen and carbon dioxide decreased along the packed bed reactor. On the other hand, the molar flow rate of water, methanol, and CO increased along the reactor. The rate of increase in the products is higher in the first 5 m of the rector but after that, it reaches a plateau. Table 4 shows the outlet molar flowrate of components. The hydrogen and CO_2 molar flowrates decreased from 1.88 to 0.58 mol/s to 1.58 and 0.47 mol/s respectively. Furthermore, the molar flowrate of methanol, carbon monoxide, and water was increased from 0, 0.07, 0 to 0.09, 0.08, and, 0.11 mol/s.

Fig. 3 shows the change in temperature and pressure along the packed bed reactor. The temperature was increased along the reactor due to the presence of exothermic reactions and reached the maximum of 537.65 K, then, the temperature was dropped to 534.73 K. Also, there was a decrease in pressure from 55 bar to 54.01 bar along the packed bed reactor. Similar temperature and pressure profiles were observed in previous studies [36].

Fig. 4 provides the molar flow rate of components along the packed bed reactor without the presence of carbon monoxide in the feed gas mixture. 25 % of the feed is CO_2 and the rest of them is hydrogen. The absence of CO in feed improved a bit H_2 and CO_2 con-

Table 2
Boundary conditions of the developed model [29].

Boundary	Mass transfer	Heat Transfer	Pressure drop
z = 0	$C_{i,in}$	T_{in}	P_{in}
r = 0	$\partial C/\partial r$	$\partial T/\partial r$	Axial Symmetry
z = L	Outlet	Outlet	Outlet
r = r_1	$\partial C/\partial r$	$\frac{Ke\times T_{in}}{r_{tot}}\times\frac{\partial T}{\partial r}=U\times\left(T_{in}-T_{ref}\right)$	wall

Table 3
Simulation parameters for the model [28,29].

Parameter	Value	unit
Entrance feed temperature	498	K
Temperature of coolant	523	K
Entrance pressure of feed	55	Bar
Diameter of tube	0.058	m
Length of tube	8	m
Flow rate	100	mol/s
Number of tube	150	–
Density of packed bed	1050	kg/m^3
Inlet carbon monoxide molar fraction	0	%
Inlet carbon dioxide molar fraction	25	%
Inlet hydrogen molar fraction	75	%
H_2O molar fraction, inlet	0	%
CH_3OH molar fraction, inlet	0	%

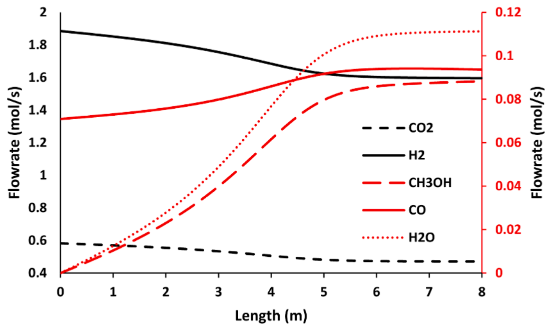

Fig. 2. The molar flow of components as a function of the length of the reactor, inlet components provided in Table 1, inlet pressure = 55 bar, inlet temperature = 498 K.

Table 4
The outlet molar flowrate of components in the current work and previous studies [28].

Gas species	Inlet Conc. (mol/s)	Outlet Conc. (mol/s)	Outlet Conc. (mol/s) [28]
Carbon monoxide	0.070	0.090	0.080
Hydrogen	1.880	1.600	1.590
Carbon dioxide	0.580	0.470	0.470
CH_3OH	0.000	0.090	0.090
H_2O	0.000	0.110	0.110

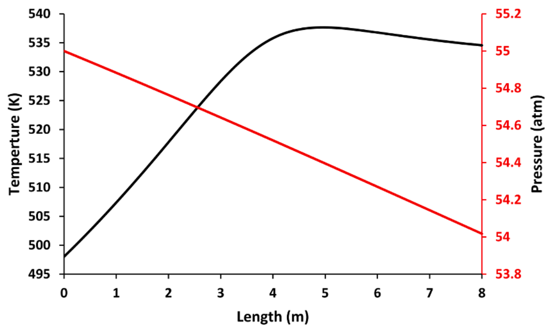

Fig. 3. Change in temperature and pressure along the packed bed reactor, inlet components provided in Table 1, inlet pressure = 55 bar, inlet temperature = 498 K.

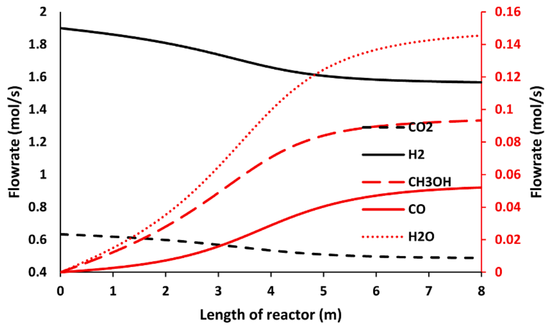

Fig. 4. Gaseous flow rates inside the methanol reactor as a function of tube length, no CO presence in the feed, CO_2 inlet flowrate = 25 %, H_2 inlet flowrate = 75 %.

sumption and enhanced the product's molar flowrate at the outlet of the reactor. The methanol content at the outlet of the gas stream was increased from 0.088 mol/s to 0.094 mol/s.

3.1. *Effect of feed inlet temperature*

Fig. 5 shows carbon dioxide profiles inside the CH_3OH reactor with various values of feed inlet temperature. The global heat exchange temperature was equal to 523 K. At the outlet of the reactor, carbon dioxide conversion is equal to 18.18 %, 22.97 %, and 23.02 %, for values of feed inlet temperature equal to 475 K, 498 K, and 555 K, respectively. It was observed that the change in feed temperature from 498 K to 555 K did not have a considerable effect on the outlet conversion of CO_2 in the reactor. However, it reached the highest value at a short distance from the entrance of the reactor. It means the shorter packed bed reactor is needed for the same conversion of CO_2. Changes in temperature along the packed bed reactor at different feed inlet temperatures have been compared in Fig. 6. Fig. 6 depicts that there is a slight increase in temperature until 4 m from the bottom of the reactor, but, after that, the rate of increase in temperature increased and it has reached 534.49 K when the inlet temperature was 475 K. At inlet temperature of 498 K, the temperature was reached a maximum of 537.65 K at distance of 5 m from entrance, then, it was decreased

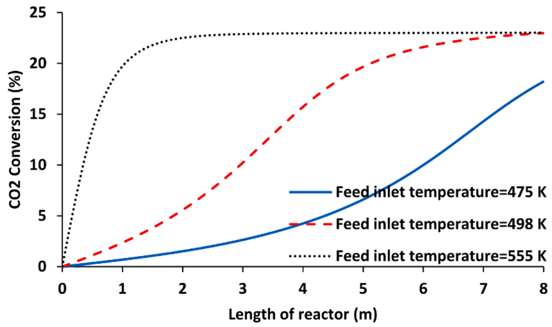

Fig. 5. CO_2 conversion as a function of reactor length at different feed inlet temperature.

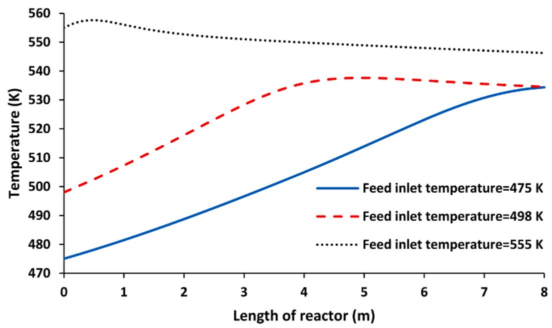

Fig. 6. Temperature profile as a function of reactor length at different feed inlet temperature.

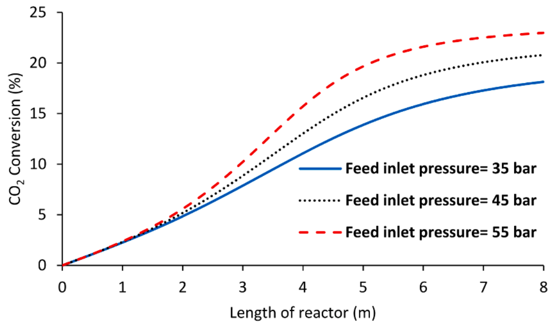

Fig. 7. CO_2 conversion as a function of reactor length at different feed inlet pressure.

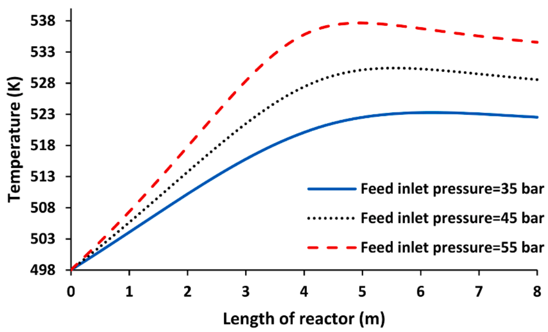

Fig. 8. Temperature profile as a function of reactor length at different feed inlet temperatures.

around 534 K at the outlet of the reactor. As shown in Fig. 6, there is a quick increase in temperature and then a gradual decrease in temperature until the end of the reactor at an inlet temperature of 555 K.

3.2. Effect of feed inlet pressure

The comparisons of carbon dioxide conversion rates along the packed bed reactor in different feed inlet pressures are presented in Fig.7. It is obvious that the rise in feed inlet pressure led to the CO_2 conversion increment. The CO_2 conversion becomes better with the increase of the inlet pressure. The conversion of CO_2 can reach up to 17.81 %, 20.52 %, and 22.81 % when the feed inlet pressure values are equal to 35 bar, 45 bar, and 55 bar respectively. Fig. 7 points out in fact that the packed bed reactor performance was improved in terms of conversion of CO_2 with the increasing of feed inlet pressure. Firstly, the increase in CO_2 conversion is linear, then there is a sharp rise especially in higher pressure, finally, the rate of increase was decreased until the end of the reactor. Fig. 8 provides temperature profiles as a function of reactor length at different pressures. The highest temperature observed in the reactor was equal to 523 K, 530 K, and 538 K at pressures of 35 bar, 45 bar, and 55 bar.

3.3. Effect of length of reactor

Fig. 9 indicates the influence of the length of the reactor on temperature profiles along the packed bed reactor. The performances of the PB reactor were affected when changing the reactor tube length. It can be seen that for all the rector lengths the same typical profile of temperature and it confirms the presence of exothermic reactions, showing a hot spot after the same distance from the reactor inlet. The temperature reaches its highest values after the same distance from the entrance for different sizes of reactor length.

4. Conclusion

To improve the process of methanol production from carbon dioxide and hydrogen and decrease its cost, a numerical simulation tool was provided in the current study. The effects of operating conditions including the presence of CO in feed, feed inlet pressure and temperature, and the length of the reactor on performance of system were systematically investigated. It was seen a decrease in H_2 and CO_2 from 1.88 to 1.58 mol/s to 0.58 and 0.47 mol/s but an increase in the molar flowrate of CH_3OH, CO, and H_2O from 0, 0.07, 0 to 0.09, 0.08, and, 0.11 mol/s respectively. The results showed that carbon dioxide conversion is equal to 18.18 %, 22.97 %, and 23.02 %, for values of feed inlet temperature equal to 475 K, 498 K, and 555 K, respectively. It was also found that the changing feed temperature from 498 K to 555 K did not have a significant influence on the outlet conversion of CO_2 in the reactor. Furthermore, the conversion of CO_2 and maximum temperature in the reactor was increased from 18.13 % to 22.97 and 523 K–538 K with

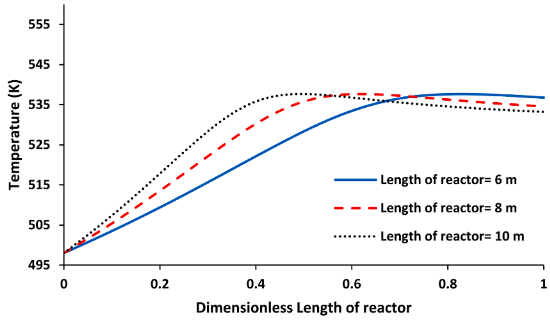

Fig. 9. Temperature profile as a function of reactor length at different lengths of the reactor.

the enhancement of feed inlet pressure from 35 bar to 55 bar. The 6 m reactor was enough for the CO_2 hydrogenation to reach its maximum temperature along the reactor.

CRediT authorship contribution statement

Jinfeng Fu: Conceptualization, Investigation, Methodology, Writing – original draft, Funding acquisition, Project administration, Supervision. **Mohammed Sh Majid:** Conceptualization, Methodology, Writing – original draft, Funding acquisition, Investigation, Project administration, Supervision. **Farag M.A. Altalbawy:** Conceptualization, Methodology, Writing – original draft, Funding acquisition, Investigation, Project administration, Supervision. **Radhwan M. Hussein:** Conceptualization, Data curation, Resources, Writing – review & editing. **Ibrahem Waleed:** Conceptualization, Data curation, Resources, Writing – review & editing. **Ibrahim Mourad Mohammed:** Conceptualization, Data curation, Resources, Writing – review & editing. **Rahman S. Zabibah:** Validation, Writing – review & editing. **Kadhum Al-Majdi:** Validation, Writing – review & editing. **Abdul Malik:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Nomenclature

2D	2-dimensional
1D	1-dimensional
A	Pre-exponential factor
CO_2	Carbon dioxide
CH_3OH	Methanol
Cp	Gas species heat capacity at constant pressure [J/(mol K)]
C_i	Molar concentration of component i
De	Effective intraporous diffusion coefficient [m^2/s]
Dr	Dispersion coefficient in r-direction
dp	Catalyst nominal diameter [m]
dt	Internal reactor tube diameter [m]
F_i	Gas species molar flow rate [mol/s]
G	Superficial flow velocity [$kgs^{-1}m^{-2}$)]
gc	Conversion factor parameter [kg/m/s^2N]
H_2	Hydrogen
Kcat	Catalyst thermal conductivity symbol [J(m s K)$^{-1}$]
Kg	Gas species thermal conductivity [kW/mK]
k_1,k_2,k_3	Reaction rate kinetic constants of methanol synthesis

$K_{p1}{}^{0}$	Equilibrium constant of reaction 1
$K_{p2}{}^{0}$	Equilibrium constant of reaction 2
$K_{p3}{}^{0}$	Equilibrium constant of reaction 3
K_{CO}	Adsorption equilibrium constant of CO (1/bar)
K_{CO2}	Adsorption equilibrium constant of CO_2 (1/bar)
K_{H2}	Adsorption equilibrium constant of H_2 (1/bar)
K_{H2O}	Adsorption equilibrium constant of H_2O (1/bar)
Ke	Thermal conductivity in radial [J/(m sK)]
Ltot	Length of reactor [m]
p	Pressure (bar)
p_{CH3OH}	Partial pressure of CH_3OH [bar]
p_{CO}	Partial pressure of CO [bar]
p_{CO2}	Partial pressure of CO_2 [bar]
p_{H2}	Partial pressure of H_2 [bar]
p_{H2O}	Partial pressure of H_2O [bar]
r_{tot}	Radius of tube in the reactor
r	Radial direction (m)
r_i	Reaction i (1–3) reaction rate [mol/(m^{3}s)]
T	Temperature (K)
Uz	Superficial gas velocity
Z	Veridical direction
ρ	Gas species density [kg m^{-3}]
ε	Packed bed reactor void fraction
μ	Viscosity of gases [Pa s]
ΔHj	reaction i (1–3) heat of reaction [kJ/mol]

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